

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
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BIOLOGY

9744/01

Paper 1 Multiple Choice

20 September 2018

1 hour

Additional Materials: OTAS Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, class and index number on the cover page and on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice **in soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

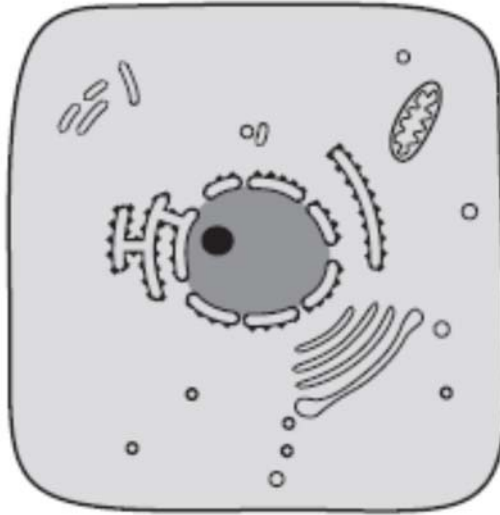
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

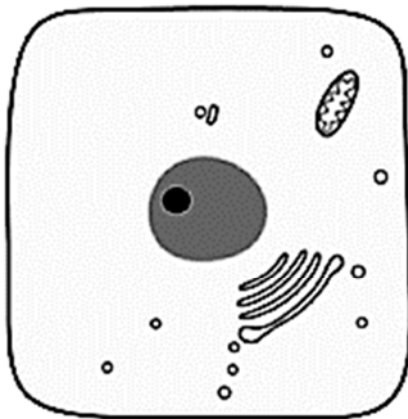
Do not open this booklet until you are told to do so.

- 1 The diagram below is drawn from an electron microscope of an animal cell.

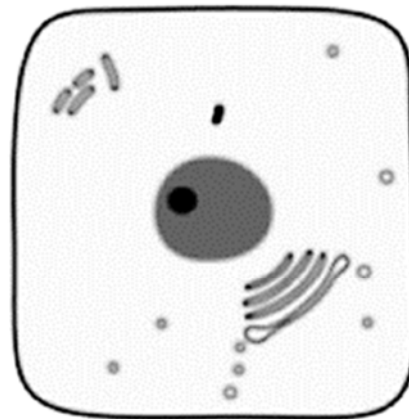


Which represents the same cell, seen under a light (optical) microscope at x400 magnification?

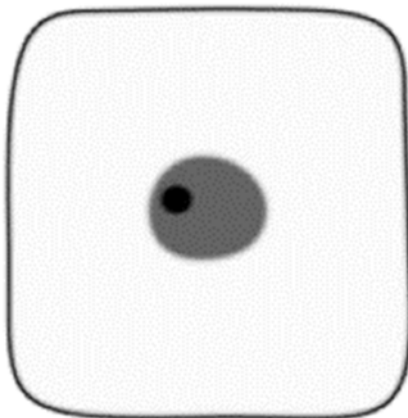
A



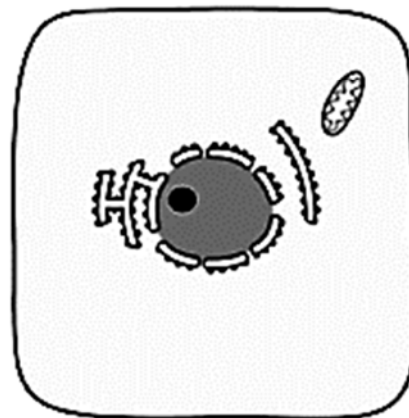
B



C



D



- 2 The diameter of living cells varies considerably. Typical diameters are:

a prokaryote, such as *Streptococcus* - 750 nm
 an eukaryotic cell, such as a white blood cell - 15 μm

Given these measurements, the diameter of the white blood cell is how many times greater than the prokaryote?

- A x 2
 B x 20
 C x 50
 D x 200
- 3 Beetroot cells contain a water-soluble red pigment. Two test tubes were set up as described in the table.

tube 1	Pieces of washed raw beetroot in water
tube 2	Pieces of washed raw beetroot in water containing 3 drops of cyanide, a respiratory inhibitor.

After 30 minutes, the water in tube 2 contained a red pigment but the water in tube 1 did not.

Which of the following statements are false for tube 2?

- I Pigment molecules passed out and were replaced by cyanide.
 II The cell membrane was unable to retain the red pigment.
 III Water entered the tissue by osmosis and caused the cells to burst.
 IV Water passed out of the cells by osmosis and carried the soluble pigment with it.
 V The same result will occur if ethanol was used instead of cyanide.
- A I and III only
 B III and IV only
 C II and V only
 D I, III and IV only

[Turn over

4 A student tested four samples of food, **A**, **B**, **C** and **D**, for the presence of

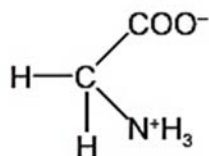
- lipids
- protein
- reducing sugars
- starch

One of the food samples, milk, was found to contain lipid, protein and reducing sugar.

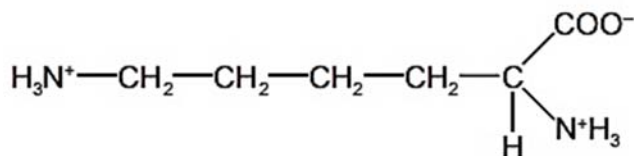
Which of the food samples, shown in the results below, is milk?

sample	observation			
	adding biuret reagent	adding iodine in potassium iodide solution	boiling with Benedict's solution	mixing with ethanol and adding to water
A	purple	orange	orange precipitate	milky emulsion
B	purple	blue-black	blue	milky emulsion
C	pale blue	blue-black	orange precipitate	clear
D	pale blue	orange	blue	clear

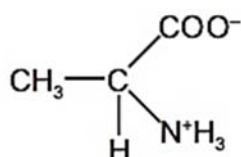
5 The diagram shows the structure of four amino acids in solution.



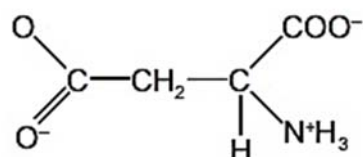
glycine



lysine



alanine

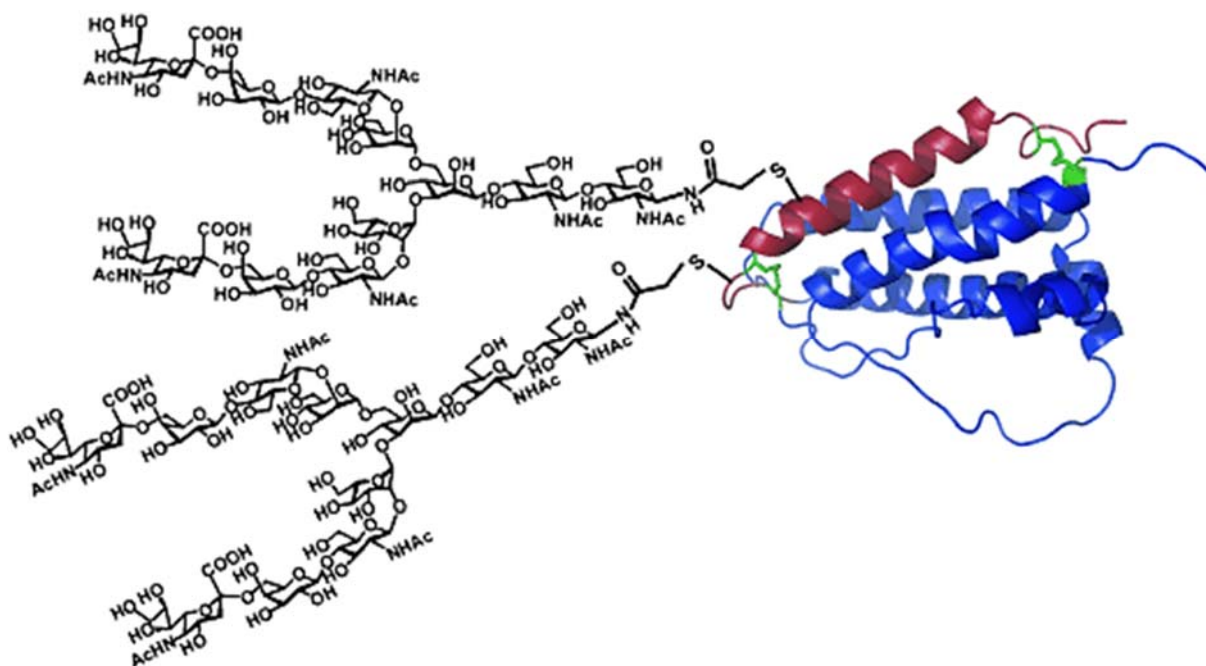


aspartate

Which of these four amino acids have an overall charge?

- A** alanine and aspartate
B alanine and glycine
C aspartate and lysine
D glycine and lysine

- 6 The figure below shows the structure of a biomolecule extracted from a cell.



Below are some statements regarding the structure, property and function of biomolecules with structures similar to that shown above. Which of the following statements are true?

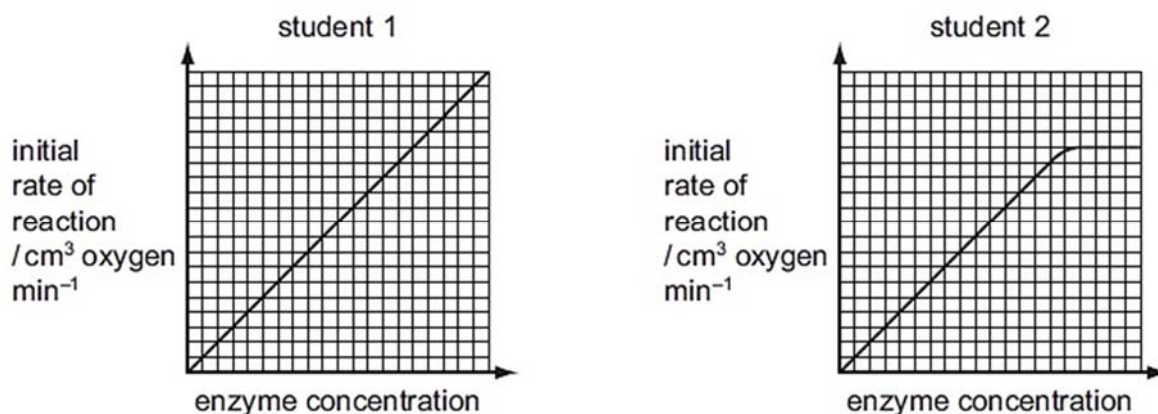
- 1 This biomolecule has both hydrophilic and hydrophobic properties.
- 2 This kind of biomolecule plays a role in blood group determination.
- 3 This biomolecule is contained within the secretory vesicle.
- 4 When completely hydrolysed, all the monomers of this biomolecule are soluble in water.

- A** 1 and 3 only
B 2 and 4 only
C 1, 2 and 4 only
D 2, 3 and 4 only

[Turn over

- 7 Catalase is an enzyme that catalyses the conversion of hydrogen peroxide into water and oxygen.

Two students investigated the effect of enzyme concentration on the rate of reaction of the enzyme catalase. The students predicted their results would show the same trend. The graphs show the rates obtained by each student.



Which statement explains the different trend shown by student 2's results?

- A Student 2 included a competitive inhibitor in the investigation.
 - B Student 2 performed the investigation at a higher temperature.
 - C Student 2 performed the investigation at pH6 compared to pH8.
 - D Student 2 used a lower concentration of substrate in the investigation.
- 8 Ethylene glycol is a chemical used to prevent water from freezing. If ethylene glycol is swallowed accidentally, it is metabolised by an enzyme found in liver cells to produce a toxic product.

The enzyme normally catalyses the oxidation of ethanol to a harmless product.

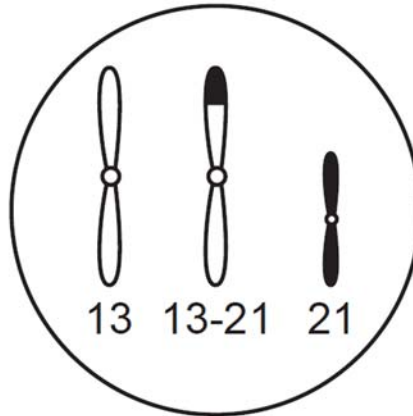
People who have swallowed ethylene glycol are treated with large doses of ethanol. This prevents formation of a toxic product and allows the body to excrete the ethylene glycol

Which statements describe why this treatment works?

- A Ethanol binds near the active site on the enzyme, altering its shape.
- B Ethanol binds permanently to the active site of the enzyme, blocking it.
- C Ethanol changes the tertiary structure of the enzyme, denaturing it.
- D Ethanol is more likely to bind to the active site on the enzyme.

- 9 Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from translocation of chromosome 21 onto chromosome 13, forming a single chromosome 13-21.

The diagram below shows chromosomes 13 and 21 in the nucleus of a diploid (2n) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.



Which is not a likely outcome of fertilisation of normal oocytes by sperm from this male?

	chromosomes in sperm	embryo
A	13 and 21	$2n = 46$ normal phenotype
B	13-21	$2n = 45$ normal phenotype
C	13-21 and 21	$2n = 46$ Down's syndrome
D	13-21 and 21	$2n = 47$ Down's syndrome

- 10 Nocodazole is a chemical used in the study of mitosis. It causes all mitotic cells to stop dividing at metaphase.

Which statements correctly identify how this chemical might work?

- 1 inhibits chromatin condensing in the nucleus
- 2 prevents replication of the centrioles
- 3 stops sister chromatids from migrating to opposite poles.

- A 1, 2 and 3
 B 1 and 2 only
 C 1 and 3 only
 D 3 only

[Turn over

11 Some of the processes of protein synthesis are listed.

- 1 condensation of amino acids
- 2 positioning of adjacent amino acids
- 3 termination of polypeptide chains
- 4 activation of amino acids
- 5 binding of amino acids to tRNA

Which processes only occur in ribosomes?

- A** 1, 2 and 3
- B** 1, 2 and 5
- C** 2, 3 and 4
- D** 3, 4 and 5

12 During semi-conservative replication of DNA in eukaryotic cells, the following processes occur.

- 1 Free nucleotides are hydrogen bonded to those on the exposed strand.
- 2 Hydrogen bonds are broken between the complementary base pairs.
- 3 The cell receives the signal to begin to divide.
- 4 Covalent bonds form between adjacent nucleotides on the same strand.
- 5 The DNA double helix is unwound.

Which shows the correct order of some of the processes?

- A** 3 → 1 → 2 → 4
- B** 3 → 2 → 4 → 5
- C** 5 → 2 → 1 → 4
- D** 5 → 2 → 3 → 1

13 At the start of the polymerase chain reaction (PCR), single stranded primers are added to the denatured DNA and the mixture cooled to 60°C.

What explains why the denatured DNA strands anneal with primers and **not** each other?

- A** The primers anneal only to the 3' end of the denatured DNA.
- B** The primers are shorter and anneal more easily.
- C** The primer concentration exceeds the denatured DNA concentration.
- D** The temperature prevents the denatured DNA from annealing together.

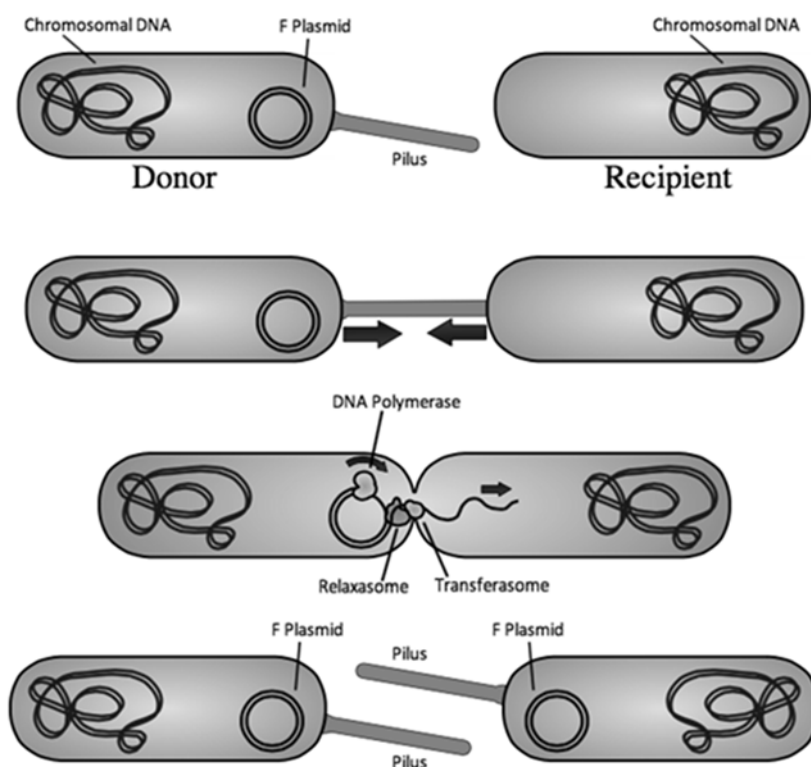
- 14** Different tissues in a plant were supplied with a radioactively labelled substance to identify which tissues were actively synthesising mRNA.

A polypeptide has the amino acid sequence glycine - arginine - lysine - serine. The table shows the possible tRNA anticodons for each amino acid.

amino acid	tRNA anticodons
arginine	UCC GCG
glycine	CCA CCU
lysine	UUC UUU
serine	AGG UCG

Which sequence of bases on the DNA would code for the polypeptide?

- A** CCA CGC AAG AGC
B CCT TCC TTC TCG
C GGA AGG AAA AGC
D GGT TGG TTG TGC
- 15** The process of bacterial conjugation is illustrated below.



[Turn over

Which of the properties of the F plasmid is **not** demonstrated in above illustration?

- A The F plasmid carries the genes encoding the pilus proteins.
- B The F plasmid has an origin of transfer.
- C The F plasmid is heritable.
- D The F plasmid replicates semi-conservatively.

16 Which of the following mutations will bring about active transcription of the lac operon in *E. coli*?

- A A mutation in the promoter that decreases the affinity of the repressor.
- B A mutation in the repressor gene that strengthens the affinity of the repressor for the inducer.
- C A mutation in the repressor gene that weakens the affinity of the repressor for the operator.
- D All of the above

17 The influenza virus enters a host cell by endocytosis and new viruses emerge by budding.

Some events that take place during the reproductive cycle are listed.

- 1 The virus enters the cytoplasm surrounded by a coated vesicle.
- 2 The nucleocapsid is released and viral RNA replicated by RNA polymerase.
- 3 Each replicated viral RNA is given a new envelope.
- 4 Viral hemagglutinin proteins bind to receptors on the host cell surface membrane.
- 5 Hemagglutinin and neuraminidase are produced by the host cell's ribosomes.
- 6 The replicated viral RNA is transcribed to mRNA.

Which sequence of events correctly describes the reproductive cycle of the virus?

- A 1 → 2 → 3 → 6 → 5 → 4
- B 1 → 2 → 6 → 5 → 4 → 3
- C 4 → 2 → 5 → 1 → 6 → 3
- D 4 → 1 → 2 → 6 → 5 → 3

18 Which of the following observations support the development of cancer as a multi-step process?

- 1 Cancer involves the accumulation of mutations in at least one proto-oncogene and several tumour suppressor genes in a specific order.
- 2 A lag time often separates the exposure to a cancer-causing agent and the development of cancer.
- 3 The incidence of cancer increases with age.
- 4 The telomerase gene is activated in malignant tumours.

- A** 1 and 3 only
B 1, 2 and 4
C 2 and 3 only
D 2 and 4 only

19 Calculation of chi-square on the results of a monohybrid cross gives a chi-squared value of 4.78. The table below shows the chi-square values.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.60	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46

What is the probability that chance produced the difference between the observed and expected?

- A** between 0.01 and 0.05
B between 0.05 and 0.1
C between 0.1 and 0.5
D more than 0.5

[Turn over

- 20** In fruit flies a sex-linked gene controls the development of eye colour. The eyes are either red or white. The male is the heterogametic sex.

What will be the expected percentages of eye colours in the progeny when a heterozygous red-eyed female is crossed with a white-eyed male?

	red eyes		white eyes	
	males	females	males	females
A	25.0	25.0	25.0	25.0
B	37.5	37.5	12.5	12.5
C	12.5	12.5	37.5	37.5
D	0	50.0	50.0	0

- 21** The table shows the occurrence of ATP production, NAD reduction and decarboxylation in different stages of aerobic respiration.

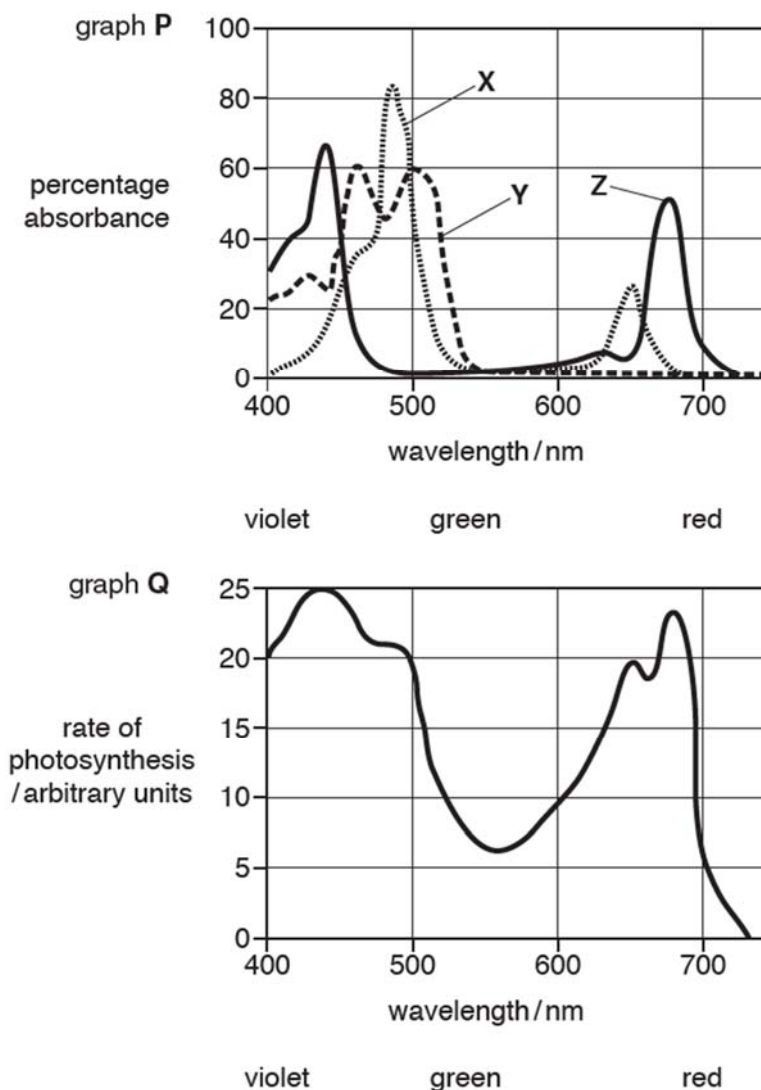
Which row is **not** correct?

	stage	ATP production	NAD reduction	decarboxylation
A	alpha-ketoglutarate to oxaloacetate	yes	yes	yes
B	citrate to alpha-ketoglutarate	no	no	yes
C	oxaloacetate to citrate	no	no	no
D	pyruvate to acetyl coenzyme A	no	yes	yes

- 22** Graph **P** shows the absorption spectra of three types of photosynthetic pigment, **X**, **Y** and **Z**, extracted from the leaves of a flowering plant.

X is chlorophyll b.

Graph **Q** shows the action spectrum for photosynthesis for the same plant.



Five students were asked to relate the information shown in graphs **P** and **Q** to their knowledge and understanding of the light-dependent stage of photosynthesis.

student	comment
1	The high absorption of blue light by chlorophyll b provides evidence that this is the primary electron donor of photosystem 1.
2	The low rate of photosynthesis in green light suggests that more green light is reflected than absorbed by the three pigments.
3	The poor absorption of green light by all three pigment types will provide only enough energy for cyclic photophosphorylation to occur.
4	Pigment Y extends the ability of the plant to absorb light in the blue-green part of the spectrum but not the yellow-green part of the spectrum.
5	Non-cyclic photophosphorylation occurs at a wavelength of 700 nm, indicating that pigment Y is more likely to be chlorophyll a than pigment Z.

[Turn over

Which students made biologically correct comments?

- A 1 and 3
- B 1 and 4
- C 2 and 4
- D 2 and 5

- 23 Different types of reaction occur in the sequence of chemical reactions known as the Calvin cycle.

- 1 Carboxylation occurs in the conversion of triose phosphate to RuBP.
- 2 Decarboxylation occurs in the conversion of RuBP to GP.
- 3 Phosphorylation occurs in the conversion of RuBP to GP.
- 4 Reduction occurs in the conversion of GP to triose phosphate.

Which of the above reactions of the Calvin cycle are **incorrectly** described?

- A 1 and 2 only
 - B 1, 2 and 3
 - C 2 and 4 only
 - D 3 and 4 only
- 24 Insulin binds to a receptor on a cell surface membrane and, as a result, the activity of an enzyme, E, in the cell is altered by phosphorylation.

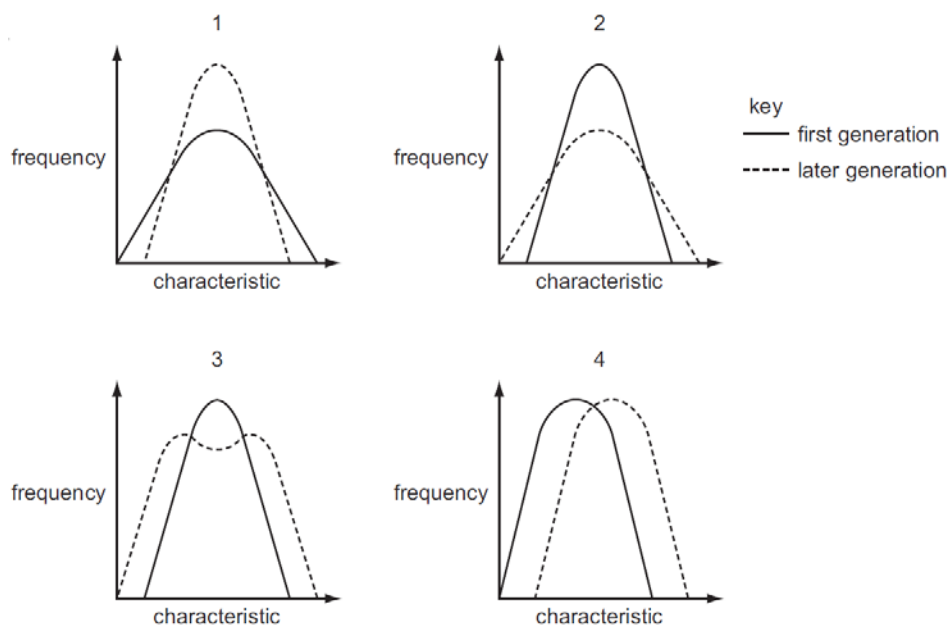
Some statements about this example of cell signalling are listed.

- 1 The concentration of cyclic AMP (cAMP) in the cell increases.
- 2 A kinase enzyme adds a phosphate group to its substrate.
- 3 Cyclic AMP (cAMP) activates an enzyme.
- 4 The enzyme adenyl cyclase is activated.

Which sequence of steps occurs to alter the activity of enzyme E?

- A $1 \rightarrow 4 \rightarrow 2 \rightarrow 3$
- B $2 \rightarrow 3 \rightarrow 4 \rightarrow 1$
- C $3 \rightarrow 2 \rightarrow 1 \rightarrow 4$
- D $4 \rightarrow 1 \rightarrow 3 \rightarrow 2$

- 25 The graphs show frequency against a measured characteristic in the first and later generation of an organism.



Which graph represents each type of natural selection?

	directional	disruptive	stabilising
A	1	2	3
B	2	3	4
C	3	1	2
D	4	3	1

- 26 The classification table below is incomplete. In addition, the taxa are not in the correct hierarchical order.

taxon	example
P	Homo
phylum	Chordata
Q	Hominidae
kingdom	R
species	Homo sapiens
S	Primates
class	T

[Turn over

Which row correctly completes the classification table?

	P	Q	R	S	T
A	genus	family	Animalia	order	Mammalia
B	genus	order	Animalia	family	Mammalia
C	order	family	Mammalia	genus	Animalia
D	order	genus	Mammalia	family	Animalia

27 Strains of *Mycobacterium* have been found that are:

- multiple drug-resistant (MDR) – resistant to the drugs most commonly used to control tuberculosis (TB)
- extensively drug-resistant (XDR) – resistant to the drugs most commonly used to control TB and to some of the drugs less commonly used to control TB
- totally drug-resistant (TDR) – resistant to all known drugs used to control TB.

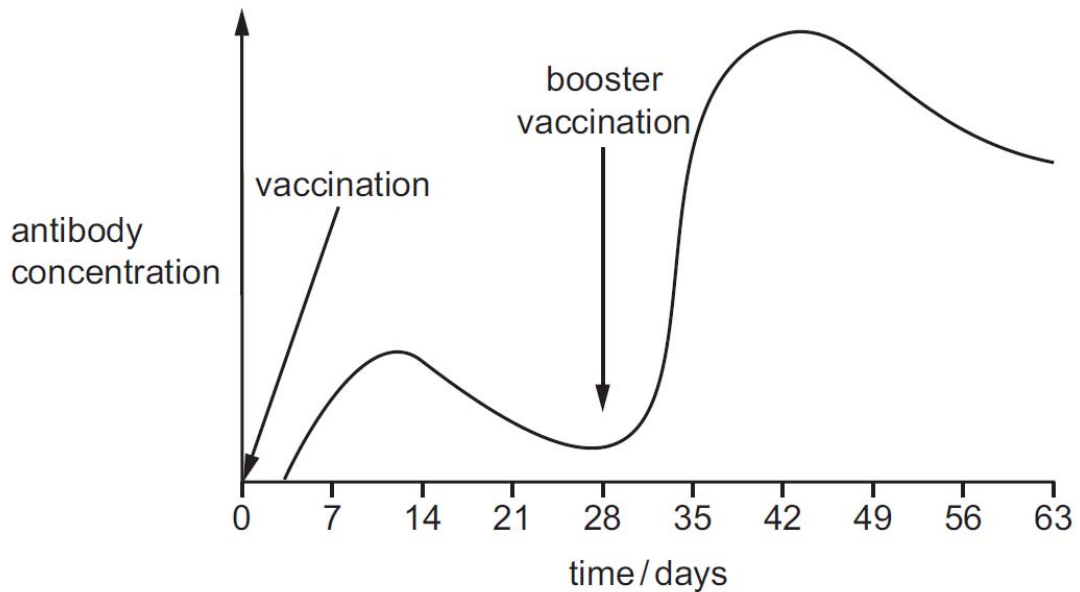
Comparisons of some of these strains of *Mycobacterium* found differences in the thickness of their cell walls, as shown in the table.

Mycobacterium strain	thickness of cell wall / nm
non-resistant	15
MDR	17
TDR	20

What conclusion may be drawn from this information?

- A** Bacteria secrete thicker cell walls when in contact with a mixture of drugs.
- B** The cell walls of TDR bacteria are impermeable to drugs.
- C** Thicker cell walls may form a physical barrier to drugs.
- D** XDR bacteria have cell walls between 17 and 20 nm thick.

- 28 The graph shows the antibody concentration in blood following vaccination and a booster vaccination 28 days later.



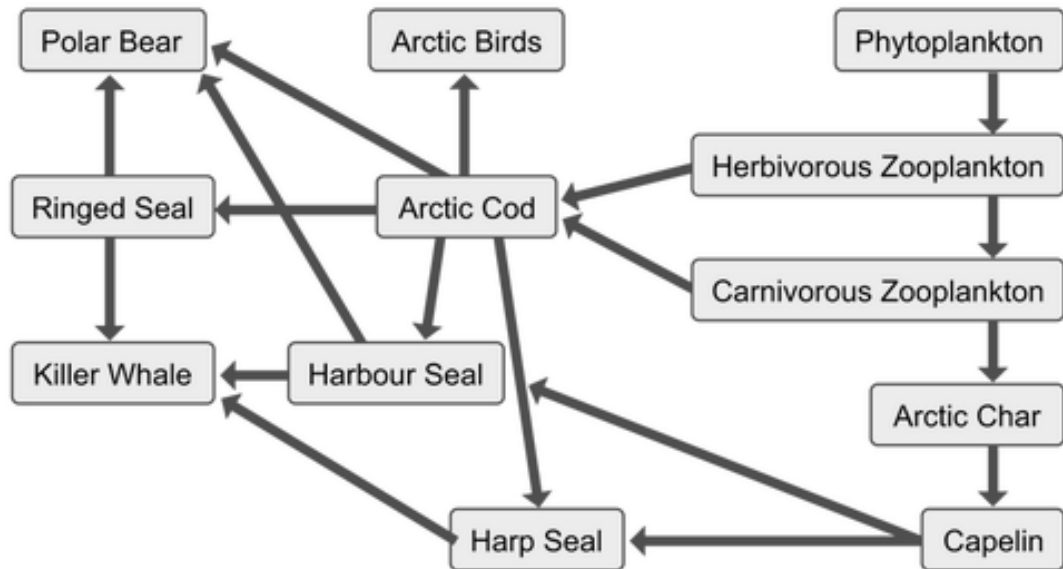
Which statements about the changes in antibody concentration are correct?

- 1 Antibody concentration falls after the primary response because antibodies are broken down and are no longer being produced.
- 2 The secondary response is more rapid due to memory B cells produced from activated B cells in the primary response.
- 3 The secondary response lasts longer than the primary response because memory B cells live longer than plasma B cells.

- A** 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

[Turn over

- 29 The Arctic Ocean is a habitat for a great number of species. The diagram below shows a simplified representation of the food web in the Arctic Ocean.



The loss of species from ecosystems as a result of anthropogenic climate change is likely to affect food webs. However, ascertaining how the removal of one species from a food web might affect others is a challenge.

Which of the following statements explain why it might be difficult to ascertain such effects?

- 1 The loss of one species might affect multiple connections in food web.
- 2 Organisms can switch their diet when their primary food source is scarce.
- 3 The consequences on a food web might take a long time to occur.
- 4 It is difficult to identify trophic levels in a food chain because of the diverse feeding behaviours.

- A** 2 and 4 only
B 3 and 4 only
C 1, 2 and 3
D 1, 2, 3 and 4

- 30** The emission of greenhouse gases such as CO₂ and CH₄ have different impacts on global warming depending on its concentration in the atmosphere, its atmospheric lifetime and its global-warming potential (GWP).

Which row correctly best describes the characteristics of these greenhouse gases?

	CO ₂	CH ₄
A	higher GWP	lower GWP
B	from anthropogenic activities such as decomposition of waste at landfills	from anthropogenic activities such as manure management of livestock
C	from natural processes such as soil respiration	from natural processes such as ruminant digestion
D	shorter atmospheric lifetime	longer atmospheric lifetime

[Turn over

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